Product Requirements Document
AI Customer Support Assistant

Version v1.0 | Status: DRAFT | Date: 2026-11-05

1. Document Information

Document Title: Product Requirements Document for AI Customer Support Assistant

Document ID: PRD-6AEBE35F

Version: 1.0 (Draft)

Target Domain: Customer Support

Product Category: Customer Support

Status: Under Active Review

2. Executive Summary

The AI Customer Support Assistant is a cutting-edge solution designed to revolutionize the customer support experience. By leveraging advanced AI capabilities, our product aims to provide accurate and helpful responses, reduce hold times, and improve customer satisfaction. With a focus on accuracy, consistency, and data privacy, we expect the AI Customer Support Assistant to become an indispensable tool for customer support teams, leading to increased efficiency, reduced costs, and enhanced customer loyalty.

3. Product Overview

Vision
To deliver an authoritative, high-integrity system solving core operational challenges in Customer Support.

Key Capabilities
- Answer customer questions using approved knowledge-base articles
- Maintain conversation history
- Identify low-confidence answers
- Escalate complex queries to human agents

4. Problem Statement

Current customer support systems often struggle to provide accurate and relevant responses, leading to long hold times and frustrated customers. The root causes of this problem lie in the limitations of legacy chatbots, which frequently provide ungrounded or inconsistent answers. This results in a cycle of inefficiency, where customers are left waiting for extended periods, and support teams struggle to provide effective solutions. The lack of context and understanding of customer needs further exacerbates the issue, making it challenging for support teams to provide personalized and effective support.

5. Product Vision and Objectives

Core Objectives
1. Provide accurate and helpful responses
2. Reduce hold times and improve customer satisfaction
3. Reduced hold times
4. Accurate and relevant customer responses

6. Target Users and Stakeholders

Primary Stakeholders and User Roles:
- **Customer Service Representative:** Primary actor interacting with AI Customer Support Assistant capabilities.
- **Customer:** Primary actor interacting with AI Customer Support Assistant capabilities.

7. User Personas

Persona 1: Customer Service Representative
- **Responsibilities:** Handling customer inquiries and resolving issues
- **Primary Goals:**
- Resolve customer issue
- Provide excellent customer service
- **Frustrations:**
- Long wait times
- Inadequate knowledge base
- **Technical Proficiency:** Intermediate

Persona 2: Customer
- **Responsibilities:** Seeking assistance with a product or service issue
- **Primary Goals:**
- Get issue resolved
- Understand product features
- **Frustrations:**
- Difficulty navigating support website
- Unhelpful support agent
- **Technical Proficiency:** Novice

Persona 3: Customer Support Manager
- **Responsibilities:** Overseeing customer support operations and ensuring quality
- **Primary Goals:**
- Monitor support performance
- Improve customer satisfaction
- **Frustrations:**
- Inadequate training for support agents
- High volume of support requests
- **Technical Proficiency:** Advanced

8. Existing Workflow and Pain Points

Currently, customer support teams operate manually or with legacy systems, which can lead to the following pain points:
* Long hold times: Customers are left waiting for extended periods, leading to frustration and a negative experience.
* Ungrounded or inconsistent answers: Legacy chatbots provide unhelpful or irrelevant responses, making it challenging for support teams to provide effective solutions.
* Lack of context: The absence of context and understanding of customer needs makes it difficult for support teams to provide personalized and effective support.
* Inefficient use of resources: Support teams spend a significant amount of time and resources on manual tasks, taking away from more critical tasks that require human attention.

9. Proposed Solution

The AI Customer Support Assistant is designed to eliminate the friction in current customer support workflows. Our solution leverages advanced AI capabilities to provide accurate and helpful responses, maintain conversation history, identify low-confidence answers, and escalate complex queries to human agents. By automating routine tasks and providing personalized support, we aim to reduce hold times, improve customer satisfaction, and increase efficiency for support teams.

10. Product Scope

- Answer customer questions using approved knowledge-base articles
- Maintain conversation history
- Identify low-confidence answers
- Escalate complex queries to human agents
- Integrate with existing CRM systems for seamless data exchange
- Support multiple channels (text, voice, and chat) for omnichannel support

11. Out of Scope

- Integration with external customer service platforms
- Implementation of advanced sentiment analysis
- Incorporation of machine learning for predictive analytics
- Development of a mobile app for on-the-go support
- Integration with other customer support tools (e.g., ticketing systems)
- Support for multiple languages beyond English

12. Functional Requirements

ID	Title	Description	Priority	Traceable AC
FR-001	Answer customer questions using approved knowledge-base articles	**Actor:** Customer Service Representative. **Precondition:** Customer Service Representative possesses authenticated session and valid role permissions for Customer Support. **System Behavior:** The system evaluates parameters, enforces business constraints, and executes answer customer questions using approved knowledge-base articles. **Constraint:** Ensure accuracy and consistency of responses. **Expected Outcome:** Answer customer questions using approved knowledge-base articles state modification is committed and verified status is presented to Customer Service Representative.	Must Have	AC-001, AC-002
FR-002	Maintain conversation history	**Actor:** Customer. **Precondition:** Customer possesses authenticated session and valid role permissions for Customer Support. **System Behavior:** The system evaluates parameters, enforces business constraints, and executes maintain conversation history. **Constraint:** Ensure accuracy and consistency of responses. **Expected Outcome:** Maintain conversation history state modification is committed and verified status is presented to Customer.	Must Have	AC-003, AC-004
FR-003	Identify low-confidence answers	**Actor:** Customer Service Representative. **Precondition:** Customer Service Representative possesses authenticated session and valid role permissions for Customer Support. **System Behavior:** The system evaluates parameters, enforces business constraints, and executes identify low-confidence answers. **Constraint:** Ensure accuracy and consistency of responses. **Expected Outcome:** Identify low-confidence answers state modification is committed and verified status is presented to Customer Service Representative.	Must Have	AC-005, AC-006
FR-004	Escalate complex queries to human agents	**Actor:** Customer. **Precondition:** Customer possesses authenticated session and valid role permissions for Customer Support. **System Behavior:** The system evaluates parameters, enforces business constraints, and executes escalate complex queries to human agents. **Constraint:** Ensure accuracy and consistency of responses. **Expected Outcome:** Escalate complex queries to human agents state modification is committed and verified status is presented to Customer.	Must Have	AC-007, AC-008

13. Non-Functional Requirements

ID	Title	Description	Priority
NFR-001	Customer Support Operational Latency	The system shall maintain 95th percentile response times under 2000ms for Customer Support transactions under baseline user concurrency. (Target bound pending formal load benchmark).	Must Have
NFR-002	Customer Support Data Protection & Security	All Customer Support session records and domain entities must be encrypted in transit and at rest. Specific regulatory standards (e.g., GDPR, FERPA, HIPAA) to be determined by institutional compliance officer review.	Must Have
NFR-003	Customer Support Availability & Fault Tolerance	The system shall achieve 99.5% availability during scheduled operational hours with automated error recovery and zero transactional data loss upon network interruption.	Must Have

14. User Stories

Story ID	As a...	I want to...	So that...	Linked FR
US-001	Customer Service Representative	answer customer questions using approved knowledge-base articles	Answer customer questions using approved knowledge-base articles state modification is committed and verified status is presented to Customer Service Representative	FR-001
US-002	Customer	maintain conversation history	Maintain conversation history state modification is committed and verified status is presented to Customer	FR-002
US-003	Customer Support Manager	identify low-confidence answers	Identify low-confidence answers state modification is committed and verified status is presented to Customer Service Representative	FR-003
US-004	Customer Service Representative	escalate complex queries to human agents	Escalate complex queries to human agents state modification is committed and verified status is presented to Customer	FR-004

15. User Journeys

```mermaid
sequenceDiagram
actor CustomerServiceRepresentative as Customer Service Representative
actor Customer as Customer
actor CustomerSupportManager as Customer Support Manager
participant AppSystem as Customer Support
Customer->>AppSystem: Searches for support resources on the website
AppSystem-->>Customer: Displays search results with relevant support articles
Note over AppSystem: Edge Case: If search query is invalid or connection drops, display
CustomerServiceRepresentative->>AppSystem: Reviews customer inquiry and identifies key issue detai
AppSystem-->>CustomerServiceRepresentative: Generates a unique ticket number and displays relevant
Note over AppSystem: Edge Case: If issue details are incomplete or invalid, prompts use
CustomerServiceRepresentative->>AppSystem: Resolves issue or escalates to supervisor for further a
AppSystem-->>CustomerServiceRepresentative: Updates ticket status and provides resolution or next s
Note over AppSystem: Edge Case: If resolution is not possible, sends automated email to
```

Step-by-Step User Journey Progression
Stage 1: 1. Discovery / Onboarding / Initial Action
- **Actor (Customer):** Searches for support resources on the website
- **System Response:** Displays search results with relevant support articles and contact options
- **Edge Case / Error Handling:** If search query is invalid or connection drops, displays error message 'Sorry, we couldn't find what you're looking for. Please try again later.'

Stage 2: 2. Issue Identification / Request Submission
- **Actor (Customer Service Representative):** Reviews customer inquiry and identifies key issue details
- **System Response:** Generates a unique ticket number and displays relevant issue details
- **Edge Case / Error Handling:** If issue details are incomplete or invalid, prompts user to provide additional information

Stage 3: 3. Resolution / Follow-up
- **Actor (Customer Service Representative):** Resolves issue or escalates to supervisor for further assistance
- **System Response:** Updates ticket status and provides resolution or next steps
- **Edge Case / Error Handling:** If resolution is not possible, sends automated email to customer with next steps and estimated resolution time

16. Business Rules

ID	Rule Title	Operational Statement	Provenance	Source Reference	Rationale	Confirmation Required
BR-001	Hold Time Threshold	Hold times for customer support should not exceed 30 minutes.	`confirmed_user_input`	75% attendance threshold	To ensure timely resolution of customer issues and maintain customer satisfaction.	No (Confirmed)
BR-002	Answer Accuracy	Customer support responses should be accurate and relevant to the customer's query.	`ai_proposed_assumption`	Industry standard for customer support best practices	To ensure customer trust and satisfaction with the support provided.	Yes (Review needed)
BR-003	Data Privacy and Security	Customer support data should be stored and processed in accordance with data privacy and security policies.	`confirmed_user_input`	Explicit data privacy and security policies	To protect customer data and maintain confidentiality.	No (Confirmed)
BR-004	Escalation Policy	Customer support issues should be escalated to a higher-level support agent if the response time exceeds 10 minutes.	`confirmed_user_input`	80% confidence escalation policy	To ensure timely resolution of critical customer issues.	No (Confirmed)
BR-005	Response Format	Customer support responses should be formatted in a clear and concise manner.	`ai_proposed_assumption`	Industry standard for customer support best practices	To ensure customer understanding and satisfaction with the support provided.	Yes (Review needed)

17. System Workflows

1. **Initial Query:** User submits a query to the AI Customer Support Assistant
2. **Conversation History:** AI maintains a conversation history to track user interactions
3. **Low-Confidence Answer:** AI identifies low-confidence answers and escalates to human agents
4. **Resolution:** AI resolves the query and updates the conversation history

18. Data Requirements

- **Core Entities:** User, Query, Answer, Conversation History
- **Attributes:** User ID, Query Text, Answer Text, Conversation History ID
- **Data Consistency Rules:** Ensure accurate and consistent responses, Maintain data privacy and security
- **Persistence Requirements:** Store conversation history in a database, Store knowledge base articles in a database

19. Integration Requirements

- Integrate with CRM system for user data
- Integrate with ticketing system for query tracking

20. Security and Privacy Requirements

- **Role Based Access Control:** Admin role, User role
- **Data Encryption:** Encrypt conversation history and knowledge base articles
- **Audit Logging:** Log all user interactions and query responses
- **Domain Specific Privacy Controls:** Comply with GDPR and CCPA regulations

21. Acceptance Criteria

AC ID	Associated Requirement	Acceptance Criteria (Given-When-Then / Rule)
AC-001	FR-001	Given Customer Service Representative initiates answer customer questions using approved knowledge-base articles with valid parameters and met prerequisites, When the request is submitted to the system, Then the transaction is processed successfully, domain records are updated, and confirmation is returned.
AC-002	FR-001	Given an edge condition or constraint breach during answer customer questions using approved knowledge-base articles (such as duplicate submission or boundary limit), When the system evaluates the transaction, Then execution is safely halted, record consistency is maintained, and a descriptive error is logged.
AC-003	FR-002	Given Customer initiates maintain conversation history with valid parameters and met prerequisites, When the request is submitted to the system, Then the transaction is processed successfully, domain records are updated, and confirmation is returned.
AC-004	FR-002	Given an edge condition or constraint breach during maintain conversation history (such as duplicate submission or boundary limit), When the system evaluates the transaction, Then execution is safely halted, record consistency is maintained, and a descriptive error is logged.
AC-005	FR-003	Given Customer Service Representative initiates identify low-confidence answers with valid parameters and met prerequisites, When the request is submitted to the system, Then the transaction is processed successfully, domain records are updated, and confirmation is returned.
AC-006	FR-003	Given an edge condition or constraint breach during identify low-confidence answers (such as duplicate submission or boundary limit), When the system evaluates the transaction, Then execution is safely halted, record consistency is maintained, and a descriptive error is logged.
AC-007	FR-004	Given Customer initiates escalate complex queries to human agents with valid parameters and met prerequisites, When the request is submitted to the system, Then the transaction is processed successfully, domain records are updated, and confirmation is returned.
AC-008	FR-004	Given an edge condition or constraint breach during escalate complex queries to human agents (such as duplicate submission or boundary limit), When the system evaluates the transaction, Then execution is safely halted, record consistency is maintained, and a descriptive error is logged.

22. Dependencies and Constraints

- Platform: Cloud-based infrastructure
- Hosting: Managed hosting service
- Operating: Linux-based operating system
- Institutional: Comply with institutional data privacy policies

23. Risks and Mitigation

Risk ID	Category	Risk Description	Root Cause	Likelihood	Impact	Mitigation Strategy	Contingency Plan	Owner
RSK-001	Operational	A customer submits a query that is not accurately answered by the knowledge base article, leading to a prolonged conversation and potential escalation to human agents.	Knowledge base article not regularly updated or reviewed by subject matter experts	Medium	Medium	Implement a regular review process for knowledge base articles, ensuring that they are accurate and up-to-date. This can be achieved by assigning a dedicated subject matter expert to review and update articles on a quarterly basis.	Have a plan in place for handling inaccurate knowledge base articles, such as providing a clear explanation to the customer and offering an alternative solution. This can be achieved by developing a standard response template for such situations.	Subject Matter Expert
RSK-002	Data Privacy	A customer's sensitive information is accessed by an unauthorized party due to insufficient encryption of conversation history data.	Insufficient encryption of conversation history data	Low	Critical	Implement end-to-end encryption for conversation history data, using industry-standard encryption protocols such as TLS. This will ensure that even if the data is accessed by an unauthorized party, it will be unreadable.	Have a plan in place for responding to a data breach, such as notifying affected customers and providing them with support. This can be achieved by developing a data breach response plan that outlines the steps to be taken in the event of a breach.	Security Engineer
RSK-003	Security	A customer submits a query that is designed to exploit a vulnerability in the database, allowing an attacker to access sensitive information.	Inadequate input validation and sanitization	Low	Critical	Implement input validation and sanitization measures to prevent SQL injection attacks. This can be achieved by using a library or framework that provides built-in protection against SQL injection.	Have a plan in place for responding to a SQL injection attack, such as immediately shutting down the affected system and notifying affected customers. This can be achieved by developing a incident response plan that outlines the steps to be taken in the event of an attack.	Security Engineer
RSK-004	Functional	A customer's query is escalated to a human agent, but the agent is not properly trained to handle the query, leading to a poor customer experience.	Inadequate training for human agents	Medium	Medium	Implement a comprehensive training program for human agents, covering a range of scenarios and query types. This can be achieved by partnering with a training provider to develop a customized training program.	Have a plan in place for handling poorly trained human agents, such as providing additional support and resources to the agent. This can be achieved by developing a support plan that outlines the steps to be taken in the event of a poorly trained agent.	Training Manager

24. Assumptions and Open Questions

Documented Operational Assumptions
- [Assumed] Inferred engineering assumptions that require validation: AI model accuracy, data quality, and scalability

Open Architectural Questions
- Critical details about customer preferences, product features, and industry-specific regulations

25. Success Metrics

- Query resolution rate
- User adoption rate
- Average response time
- Conversation closure rate

26. Release Considerations

- Phased rollout: Alpha, Beta, and Production
- Alpha validation: 30 days
- Beta validation: 60 days
- Data migration: 90 days
- Operational handoff: 120 days

27. Appendix and Glossary

- **Domain Specific Terms:** Knowledge base article, Conversation history, User interaction
- **Acronyms:** CRM: Customer Relationship Management, Ticketing system: Ticketing and support system
- **Operational Definitions:** Query response: The answer provided by the AI to the user's query, Conversation closure: The process of closing the conversation and updating the conversation history